

Mapping for a Cylindrical Micro-Resistive Well Detector

2. Group Members:

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4. Client:

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5. Dates and Meetings with Client:

1. August 28th, 2026
2. August 31st, 2026

6. Goal and Motivation:

The goal for the project is to create a program that characterises the efficiency and resolution for the cylindrical micro resistive well detector. The motivation for the project is to provide support for the software integration so the data can be more easily visualized and analyzed by the physicists.

7. Approach:

- **Update APV mapping for ROOT file representation**

The first feature is to provide a correct mapping of the APV's in the root file representation of the experiment data. Allowing for data that is useful in other parts of the pipeline. To break it down, the ROOT file is a program used to store the massive amounts of data collected by all the detectors. The cylindrical micro resistive well detector has six APV chips each with 128 channels. Each of the channels are classified two ways: globally, by the order in which they appear and locally, by the orientation of the particular APV chip. Currently, information about each channel is stored in branches in the ROOT file. Essentially, we need an efficient method for updating the channel representation without losing or affecting the existing data.

- **Automate pipeline for resolution plot generation**

- From the corrected data, another feature requested by the physicists is an automated pipeline that goes from data collection to generation of resolution plots. We will use the channel information from the updated ROOT file to reconstruct the 2D hit positions and display them in a clear visual representation. These resolution plots will use color and/or depth to illustrate the voltage levels. We expect these plots will also aid in highlighting the efficiency loss of dead channels.

Automate pipeline for efficiency plot generation

- Our next feature is another type of plot generation automation. In this case, efficiency is used to evaluate the correctness of the cylindrical detector compared to the four other detectors in identifying the hit positions. The physical quantity is the ratio of the number of events with strip clusters seen in all five detectors divided by the number of events with strip clusters seen in just the four trackers. Our pipeline will calculate the efficiency and plot it against other performance characteristics such as amplification, drift field, or other relevant factors, at the request of our clients.

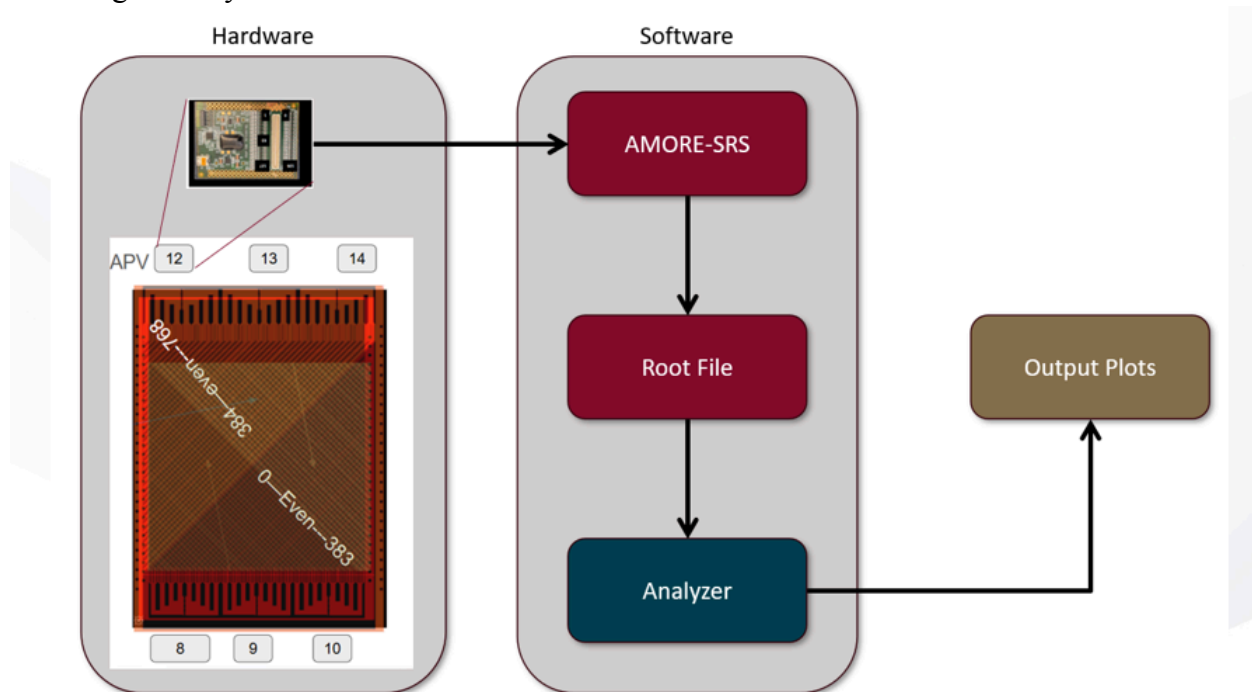
8. Novel Features:

The software development's novel features include its streamlined ability to take multiple beam data files, apply them to correct pin to physical strip mapping, visualize electron cluster location on the detector, and provide efficiency and resolution analysis.

9. Algorithms and Tools:

C++, Python, ROOT (data analysis framework for C++ and Python), AMORE-SRS tools

10. Design and System Architecture:



Our input to the analyzer is a root file consisting of event data from the APVs on the trackers and cylindrical detector. For each root file input into the stream, we will first preprocess the data, organize it and filter out junk points that would create poor results. Then we apply the mapping functions to the strips. Next, we generate the reconstruction and efficiency plots which are our output of the pipeline.

11. Evaluation:

We plan to measure the success in terms of speed, accuracy, reliability, and user friendliness. Speed is important because our input root files each contain about 2 million data points;

therefore, we have multithreaded the processing to account for this and we aim to keep the processing time under two minutes per file. We have several methods to measure the accuracy of the mapping which is the main aim of our project. Once we reconstruct a beam that looks nominal, we have verification methods for comparing the results to planar data and generating efficiency plots. For reliability, it's important that we include proper support for error handling things such as empty files or junk data points that exist outside of their expected ranges. Following our test plan will help keep the software reliable. Lastly, we are creating the software for physicists to use so it's important we ensure they understand how to use the program though we expect it to have quite a small UI for intuitive use.

12. Progress Summary:

Feature	Completion	To do
Perform reconstruction on planar data (from trackers)	100%	Nothing
Implement mathematical detector model to find intersecting strips	100%	Nothing
Generate pulse plots	100%	Nothing
Detector 5 mapping	70%	We have to find the correct mapping of 2 remaining APVs
Compare cylindrical detectors and trackers	15%	We have to invest methods for handling the offset
Create efficiency plots	0%	We haven't got the mapping to be able to create efficiency plots

13. Technical Challenges:

- APVs from the cylindrical detector seem to be not in the order that we expected.
- We have no way to access the APVs to see the orientation.

14. Milestone 4: Our first task for this semester is to find the correct mapping for the cylindrical detector.

- Do beam reconstruction with different mappings so that we can find the correct one:
- Create efficiency plots for the different mappings..
- Compare different mappings against the correct tracker data.

15. Milestone 5: Use multiple root files to test our mappings and do data analysis on each of them:

- Do beam reconstruction on multiple root files with all detectors and trackers
- Do efficiency plots for multiple root files with all detectors and trackers.
- Automating the generation of efficiency plots against other relevant performance characteristics.
- Automating the generation of resolution plots to display the positions.

16. Milestone 6: Write paper about the work done, and do a demo of the work done on the project. This milestone includes:

- Write paper.
- Create a demo video.
- Evaluate results.
- Create documentation for the CLI

17. Task Matrix Milestone 4:

Task	Alejandro	Summer
Find correct cylindrical mapping	50%	50%
Create efficiency plots	50%	50%
Compare cylindrical detectors and trackers	50%	50%

15. Approval from Faculty Advisor

- "I have discussed with the team and approve this project plan. I will evaluate the progress and assign a grade for each of the three milestones."
- Signature: _____ Date: _____